

OSI Model — Network Protocols

Layer-wise Quick Reference Guide



DATA LINK LAYER (Layer 2)

Protocol Name	Key Features / Points
Ethernet (IEEE 802.3)	➤ Most widely used wired LAN technology today ➤ Uses CSMA/CD (Carrier Sense Multiple Access / Collision Detection) to manage access ➤ Supports speeds: 10 Mbps → 10 Gbps+ ➤ Data sent in frames containing MAC addresses ➤ Defined by IEEE 802.3 standard
Token Ring	➤ Devices connected in a ring topology ➤ Uses a token (special bit pattern) to control who can transmit — no collisions ➤ Operates at 4 or 16 Mbps ➤ Largely obsolete, replaced by Ethernet ➤ Defined by IEEE 802.5 standard
PPP (Point-to-Point Protocol)	➤ Used for direct connections between two nodes (e.g., dial-up internet) ➤ Supports authentication (PAP, CHAP), compression, and encryption ➤ Works over serial links, phone lines, and fiber optic connections ➤ Encapsulates multiple network layer protocols (IP, IPX) ➤ Foundation for PPPoE (used in DSL broadband)
STP (Spanning Tree Protocol)	➤ Prevents network loops in Ethernet networks with redundant paths ➤ Elects a Root Bridge — all paths are calculated from it ➤ Automatically blocks redundant ports to eliminate loops ➤ If main path fails, STP re-enables a blocked path (failover) ➤ RSTP (802.1w) is faster modern version; MSTP supports VLANs
VLAN (Virtual LAN)	➤ Logically segments a physical network into multiple isolated networks ➤ Devices in same VLAN can communicate as if on same switch regardless of location ➤ Improves security by isolating traffic (e.g., HR vs Finance VLAN) ➤ Reduces broadcast traffic by limiting broadcast domains ➤ Defined by IEEE 802.1Q — uses VLAN tagging on frames



NETWORK LAYER (Layer 3)

Protocol Name	Key Features / Points
ARP (Address Resolution Protocol)	> Maps IP addresses → MAC addresses within a local network > Device broadcasts 'Who has IP X?' — owner replies with its MAC > Results stored in ARP cache to avoid repeated lookups > Gratuitous ARP: device announces its own IP/MAC (used in failover) > Vulnerable to ARP spoofing/poisoning attacks
RARP (Reverse ARP)	> Opposite of ARP — maps MAC address → IP address > Used by diskless workstations to discover their IP at boot time > Largely obsolete — replaced by BOOTP and then DHCP > Requires a RARP server on the local network
ICMP (Internet Control Message Protocol)	> Used for error reporting and diagnostics in IP networks > Ping uses ICMP Echo Request/Reply to test connectivity > Traceroute uses ICMP to map network path hop-by-hop > Reports errors like Destination Unreachable, Time Exceeded, Redirect > Operates at Network Layer but within IP packets (not TCP/UDP)
NAT (Network Address Translation)	> Translates private IP addresses → public IP address for internet access > Allows many devices to share a single public IP (conserves IPv4 addresses) > Provides a layer of security by hiding internal network structure > Types: Static NAT (1-to-1), Dynamic NAT (pool), PAT (many-to-one) > Commonly implemented in home routers and enterprise firewalls
PAT (Port Address Translation)	> A type of NAT — maps many private IPs to one public IP using different ports > Also called NAT Overload — most common form of NAT > Each connection tracked by IP + Port number combination > Enables thousands of devices to share one public IP simultaneously > Used in virtually all home and small business routers



SESSION LAYER (Layer 5)

Protocol Name	Key Features / Points
NetBIOS	➤ Provides session services for Windows networking ➤ Supports name registration (each device gets a NetBIOS name) ➤ Enables session establishment between applications on different machines ➤ Used in Windows file sharing and older LAN Manager networks ➤ Largely replaced by DNS in modern Windows environments
ADSP (AppleTalk Data Stream Protocol)	➤ Session-layer protocol from Apple's AppleTalk suite ➤ Provides full-duplex byte-stream communication between AppleTalk nodes ➤ Supports connection establishment, data transfer, and teardown ➤ Obsolete — AppleTalk discontinued in macOS 10.6 (Snow Leopard)
RTCP (Real-Time Transport Control Protocol)	➤ Companion to RTP — provides out-of-band statistics and control ➤ Monitors QoS (Quality of Service): packet loss, jitter, delay ➤ Sends sender/receiver reports to all participants in a session ➤ Used in VoIP, video conferencing, streaming media ➤ Helps with synchronization of multiple media streams
PPTP (Point-to-Point Tunneling Protocol)	➤ Creates VPN tunnels over TCP/IP networks ➤ Uses TCP port 1723 for control connection ➤ Encapsulates PPP packets inside IP packets for tunneling ➤ Fast but weak security — considered deprecated due to vulnerabilities ➤ Replaced by L2TP/IPsec, OpenVPN, WireGuard in modern deployments
PAP (Password Authentication Protocol)	➤ Simple username/password authentication used with PPP ➤ Credentials sent in plaintext — highly insecure ➤ No encryption of the handshake — vulnerable to eavesdropping ➤ Deprecated — replaced by CHAP (Challenge Handshake Authentication Protocol) ➤ Still seen in legacy dial-up and some embedded systems
RPC (Remote Procedure Call)	➤ Allows a program to execute a function on a remote server as if it were local ➤ Abstracts network communication — developer calls a stub function ➤ Used in NFS, Active Directory, Windows services ➤ Supports synchronous (blocking) calls by default ➤ Foundation of distributed computing and microservices architecture
gRPC (Google Remote Procedure Call)	➤ Modern, high-performance RPC framework from Google

Protocol Name	Key Features / Points
	> Uses Protocol Buffers (protobuf) for compact binary serialization > Built on HTTP/2 — supports multiplexing and streaming > 4 communication modes: unary, server streaming, client streaming, bi-directional > Widely used in microservices and cloud-native applications
SDP (Sockets Direct Protocol)	> Provides socket-level access to RDMA (Remote Direct Memory Access) networks > Bypasses the OS kernel for ultra-low latency data transfer > Designed for high-performance computing (HPC) environments > Compatible with existing BSD socket APIs — minimal code changes needed > Works over InfiniBand and other RDMA-capable networks
✓ SCP (Secure Copy Protocol)	> Used to securely transfer files between hosts over a network > Built on top of SSH — inherits its strong encryption > Simple command: scp file user@host:/path > Supports recursive copy of directories with -r flag > Being replaced by SFTP/rsync in modern usage (SCP lacks interactive features)



PRESENTATION LAYER (Layer 6)

Protocol Name	Key Features / Points
✓ SSL (Secure Sockets Layer)	> Cryptographic protocol to secure communications over a network > Provides encryption, authentication, and data integrity > Deprecated — all versions (SSL 2.0, 3.0) have known vulnerabilities > Replaced by TLS (Transport Layer Security) > Still commonly referenced when people say 'SSL certificate' (actually TLS)
✓ TLS (Transport Layer Security)	> Modern successor to SSL — the standard for securing internet communications > Provides encryption (confidentiality), authentication, and integrity > TLS 1.3 (2018) is current standard — significantly faster handshake > Used in HTTPS, email (SMTPS), VPN, and most secure protocols > Uses certificates (X.509) from Certificate Authorities (CAs) for identity
AFP (Apple Filing Protocol)	> Network protocol for file sharing on Apple/Mac networks

Protocol Name	Key Features / Points
	➤ Supports Mac-specific metadata: resource forks, long filenames, permissions ➤ Provides authentication and encryption for file access ➤ Being phased out by Apple — SMB (Server Message Block) is now preferred ➤ Operates over TCP port 548
LPP (Lightweight Presentation Protocol)	➤ Provides a subset of OSI presentation services with low overhead ➤ Designed for use with ISO applications running over TCP/IP ➤ Used to encode and decode ASN.1 data structures efficiently ➤ Rarely seen in modern networks — primarily a legacy/academic protocol
NCP (NetWare Core Protocol)	➤ Core protocol for Novell NetWare file/print sharing services ➤ Provides file access, directory services, and print queue management ➤ Handles data encoding/decoding between client and NetWare server ➤ Largely obsolete with decline of Novell NetWare
NDR (Network Data Representation)	➤ Data encoding standard used by Microsoft's DCE/RPC and DCOM ➤ Handles byte order differences between systems (little-endian vs big-endian) ➤ Converts complex data types (structs, arrays) for cross-platform transmission ➤ Integral to Windows networking — used in Active Directory, DCOM, SMB
XDR (External Data Representation)	➤ Standard for encoding data for transfer between heterogeneous systems ➤ Always uses big-endian byte order (network byte order) ➤ Defined in RFC 4506 — used with Sun RPC / ONC RPC ➤ Foundation of NFS (Network File System) data encoding ➤ Handles integers, floats, strings, arrays — platform-independent



APPLICATION LAYER (Layer 7)

7.1 Web Browsing Protocols

Protocol Name	Key Features / Points
HTTP (HyperText Transfer Protocol)	➤ Foundation of the World Wide Web — transfers web pages and data ➤ Stateless protocol — each request is independent

Protocol Name	Key Features / Points
	➤ Methods: GET, POST, PUT, DELETE, PATCH, HEAD, OPTIONS ➤ HTTP/2 (2015): multiplexing, header compression, server push ➤ HTTP/3 (2022): runs over QUIC (UDP-based) for even faster loading
HTTPS (HTTP Secure)	➤ HTTP encrypted with TLS — the secure version of HTTP ➤ Provides encryption, authentication, and data integrity ➤ Uses port 443 (HTTP uses port 80) ➤ Website identity verified via SSL/TLS certificates ➤ Mandatory for modern websites — browsers flag HTTP sites as insecure

7.2 Email Protocols

Protocol Name	Key Features / Points
SMTP (Simple Mail Transfer Protocol)	➤ Protocol for sending and relaying email between servers ➤ Uses port 25 (server-to-server), 587 (client submission), 465 (SMTPS) ➤ Push protocol — sender pushes mail to recipient's server ➤ Works with MTA (Mail Transfer Agents) like Postfix, Sendmail ➤ Combined with DKIM, SPF, DMARC to prevent spam/spoofing
POP3 (Post Office Protocol v3)	➤ Used to download email from server to client device ➤ Typically deletes mail from server after download (offline model) ➤ Uses port 110 (plain) or 995 (POP3S with TLS) ➤ Simple but limited — no folder sync, no multi-device support ➤ Being replaced by IMAP for multi-device email access
IMAP (Internet Message Access Protocol)	➤ Allows accessing and managing email directly on the server ➤ Syncs across multiple devices — mail stays on server ➤ Supports folders, flags, search on server-side ➤ Uses port 143 (plain) or 993 (IMAPS with TLS) ➤ Industry standard for modern email clients (Outlook, Gmail, Thunderbird)

7.3 File Transfer Protocols

Protocol Name	Key Features / Points
FTAM (File Transfer, Access and Management)	➤ OSI standard for file transfer and management (ISO 8571) ➤ Supports file access, transfer, and management operations ➤ More complex than FTP — designed for heterogeneous environments

Protocol Name	Key Features / Points
	> Largely obsolete in modern networks — replaced by FTP/SFTP/SCP
FTP (File Transfer Protocol)	> Standard protocol for transferring files between client and server > Uses two connections: control (port 21) and data (port 20 / passive) > Modes: Active (server initiates data) and Passive (client initiates data) > Supports authentication but transfers data in plaintext (insecure) > Use FTPS or SFTP for secure alternatives
TFTP (Trivial File Transfer Protocol)	> Simplified file transfer — no authentication, no directory listing > Uses UDP port 69 — very lightweight, minimal overhead > Used for booting devices over network (PXE boot, network switches) > Used to update firmware on routers, switches, and embedded devices > No security — only suitable for trusted local networks
SFTP (Secure File Transfer Protocol)	> Secure file transfer built on SSH — not related to FTP despite name > Provides encryption, authentication, and integrity for all transfers > Supports interactive operations: browse, rename, delete, permissions > Uses SSH port 22 — only one connection needed > Preferred choice for secure file transfers in modern environments

7.4 Network Management / Support Protocols

Protocol Name	Key Features / Points
DHCP (Dynamic Host Configuration Protocol)	> Automatically assigns IP addresses and network settings to devices > Provides: IP address, subnet mask, default gateway, DNS server > DORA process: Discover → Offer → Request → Acknowledge > Uses UDP ports 67 (server) and 68 (client) > Supports IP reservations (bind specific IP to a MAC address)
DNS (Domain Name System)	> Translates domain names → IP addresses (the internet's phone book) > Hierarchical: Root → TLD (.com, .org) → Domain → Subdomain

Protocol Name	Key Features / Points
	➤ Record types: <u>A</u> (IPv4), <u>AAAA</u> (IPv6), <u>MX</u> (mail), <u>CNAME</u> (alias), <u>TXT</u> ➤ Uses <u>UDP port 53</u> (and TCP for large responses/zone transfers) ➤ <u>DNS caching</u> reduces lookup time — TTL controls cache duration
SNMP (Simple Network Management Protocol)	➤ Used to <u>monitor and manage network devices</u> (routers, switches, printers) ➤ Components: <u>Manager</u> (NMS), <u>Agent</u> (on device), <u>MIB</u> (data definitions) ➤ Operations: <u>GET, SET, TRAP</u> (unsolicited alert from agent) ➤ Uses <u>UDP port 161</u> (queries) and <u>162</u> (traps) ➤ <u>SNMPv3</u> adds encryption and authentication — use it over v1/v2c

7.5 Remote Access Protocols

Protocol Name	Key Features / Points
Telnet	➤ Provides <u>remote command-line access</u> to another computer ➤ Uses <u>TCP port 23</u> ➤ <u>Completely insecure</u> — all data including passwords sent in plaintext ➤ <u>Deprecated</u> — replaced entirely by SSH for all remote access ➤ Still used in <u>legacy systems, IoT devices</u>, and network testing
SSH (Secure Shell)	➤ <u>Encrypted replacement for Telnet</u> — secure remote terminal access ➤ Uses <u>TCP port 22</u> — provides strong <u>encryption by default</u> ➤ Supports <u>public key authentication</u> (more secure than passwords) ➤ Enables <u>port forwarding / SSH tunneling</u> to secure other protocols ➤ Used for <u>remote admin, SFTP, SCP, Git over SSH</u>, and more

7.6 Other Application Layer Protocols

Protocol Name	Key Features / Points
NFS (Network File System)	➤ Allows <u>mounting remote file systems</u> as if local (transparent access) ➤ Developed by <u>Sun Microsystems</u> — standard in Unix/Linux environments ➤ <u>Stateless</u> (NFSv3) or <u>stateful</u> (NFSv4) protocol ➤ Uses <u>RPC and XDR</u> for communication and data encoding ➤ NFSv4 adds <u>security (Kerberos), ACLs, and better WAN performance</u>

Protocol Name	Key Features / Points
LPD (Line Printer Daemon)	➤ Network printing protocol — one of the oldest print protocols ➤ Uses TCP port 515 — client is called LPR (Line Printer Remote) ➤ Supports print job queuing on remote printers ➤ Largely replaced by IPP (Internet Printing Protocol) on modern systems ➤ Still supported in Unix/Linux printing systems (CUPS)
X Window System	➤ Graphical display protocol for Unix/Linux — enables remote GUI applications ➤ Client-server model — display server (local) renders graphics for apps (remote) ➤ Allows running graphical apps on a remote machine, displayed locally ➤ Basis of desktop environments like GNOME, KDE on Linux ➤ Uses TCP port 6000+; being succeeded by Wayland in modern Linux
IRC (Internet Relay Chat)	➤ Real-time text messaging protocol for group chat (channels) and private messages ➤ Uses TCP port 6667 (plain) or 6697 (TLS) ➤ Organized into servers, networks, channels (#channel), and users ➤ Decentralized — multiple servers can form a network ➤ Influenced modern chat apps; still used in developer/open-source communities
MIME (Multipurpose Internet Mail Extensions)	➤ Extends email to support non-ASCII content (attachments, HTML, multimedia) ➤ Defines Content-Type headers (e.g., text/html, image/jpeg, application/pdf) ➤ Enables email attachments via Base64 encoding ➤ Also used in HTTP for content negotiation (Accept, Content-Type headers) ➤ Essential to modern email — virtually all rich email uses MIME
NNTP (Network News Transfer Protocol)	➤ Protocol for reading and posting Usenet newsgroup articles ➤ Uses TCP port 119 (plain) or 563 (NNTPS) ➤ Articles organized in hierarchical newsgroups (e.g., comp.lang.python) ➤ Servers replicate articles between each other (distributed system) ➤ Largely obsolete — Usenet replaced by web forums, Reddit, Discord
LDAP (Lightweight Directory Access Protocol)	➤ Protocol for accessing and managing directory services (user databases) ➤ Used to query Microsoft Active Directory, OpenLDAP user stores ➤ Stores info in a tree structure (DIT) — entries identified by DN ➤ Operations: bind, search, add, modify, delete, compare

Protocol Name	Key Features / Points
	➤ Supports LDAPS (LDAP over TLS) on port 636; plain LDAP uses port 389
NTP (Network Time Protocol)	➤ Synchronizes clocks of computers across a network ➤ Uses a hierarchical stratum system — Stratum 0 = atomic clock, Stratum 1 = server directly connected to it ➤ Achieves millisecond-level accuracy over the internet ➤ Uses UDP port 123 ➤ Critical for SSL certificates, Kerberos, log analysis, and distributed systems
MQTT (Message Queue Telemetry Transport)	➤ Lightweight pub/sub messaging protocol designed for constrained devices ➤ Ideal for IoT (Internet of Things) — sensors, smart home, industrial monitoring ➤ Uses a broker (e.g., Mosquitto) to route messages between publishers and subscribers ➤ Uses TCP port 1883 (plain) or 8883 (TLS) ➤ QoS levels: 0 (fire and forget), 1 (at least once), 2 (exactly once)
SIP (Session Initiation Protocol)	➤ Signaling protocol for initiating, managing, and terminating multimedia sessions ➤ Used in VoIP calls, video conferencing, instant messaging ➤ Text-based protocol similar in structure to HTTP ➤ Uses UDP/TCP port 5060 (plain) or 5061 (TLS) ➤ Works alongside RTP (for actual media) and SDP (for session negotiation)